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Cyber Defense Club Notes
31-Oct-2026

Proxmox VM Performance And Hardware Configuration:

Chris explained how the team can improve VM performance by changing CPU and display settings, using VirtViewer, and manually migrating VMs between Proxmox hosts when needed.

- **CPU Type Selection:** Chris instructed participants to shut down a VM, open Hardware, select Processors, and use the x86-64-v2-AES CPU type rather than the Host type. Host exposes all processor capabilities but prevents migration between nodes; x86-64-v2-AES preserves most capabilities while allowing VMs to migrate and may improve performance.
- **VM Migration:** Chris observed that many VMs had initially been placed on host-03, contributing to slow performance. If a VM remains on host-03 or runs slowly, participants can use Proxmox's Migrate action to move it to a less-used host; the migration generally completes quickly.
- **VirtViewer Installation:** For Windows, Chris demonstrated searching for and installing VirtViewer with Winget using the Red Hat VirtViewer package. On Ubuntu, participants should run sudo apt update followed by sudo apt install virt-viewer. Chris clarified that VirtViewer is installed on the participant's own computer, not inside the VM.
- **SPICE Display Use:** Chris explained that SPICE can be enlarged to a normal full-screen view, automatically resize the guest display, expose keyboard shortcuts, and provide optional sound routing. If the desktop is too small, participants can adjust display scaling, although fractional scaling may reduce performance or sharpness.

Ubuntu Networking And Proxy Configuration:

Chris worked with participants to correct VM networking, replace incorrect DHCP or private-network settings with the club's static network configuration, and configure the proxy required for package downloads.

- **Static Network Settings:** Chris identified that a VM showing a 192.168.x.x address was on the wrong network. The intended configuration uses 144.76.90.<VM number> with a 255.255.255.0 netmask, gateway 144.76.90.254, and DNS server 199.100.16.100. He cautioned against entering slash notation in the graphical address field because it caused the configuration to fail.
- **Network Interface Troubleshooting:** When a participant's connection remained unavailable, Chris used ip -br a and ip a to inspect interfaces, checked the MAC address and NetworkManager profiles, removed the profile associated with the 192.168.x.x address, enabled the correct profile, and used sudo ip link set ens18 up. Rebooting was suggested after applying the settings.
- **Proxy Configuration:** Chris directed participants to use the documented proxy at 199.100.16.100 with port 3128. The proxy applies to the relevant HTTP, HTTPS, FTP, and SOCKS settings; the Ubuntu APT configuration also needs to be updated so apt, curl, and wget can access external repositories.
- **Network Verification:** Participants can verify the result with ip a or ip -br a. A 144.76.90.<VM number> address indicates the intended network, while a 192.168.x.x address indicates that the VM is still using the wrong network.

SSH Server Installation And Validation:

Chris guided the group through installing, enabling, testing, and understanding OpenSSH on Ubuntu VMs, including first-time host-key verification and remote login testing.

- **OpenSSH Installation:** Chris prescribed sudo apt update followed by sudo apt install openssh-server. Some VMs already had the package installed, while others were delayed by network, repository, or proxy problems.
- **Service Status:** Chris explained that installing OpenSSH does not necessarily mean the service is enabled and running automatically. Participants should use systemctl status ssh or systemctl status sshd and confirm that the service is active and enabled; active means it is currently running, while enabled means it will start after reboot.

- **SSH Connection Syntax:** To connect, participants use ssh username@IP-address. Because name resolution is not yet configured, the current tests use IP addresses. Chris successfully demonstrated connecting to another VM after entering the correct password.
- **Host-Key Prompt:** On the first connection to a host, SSH displays an authenticity warning and an ED25519 fingerprint. Chris explained that accepting the prompt records the host identity in the user's hidden ~/.ssh/known_hosts file; connecting to the same machine under a different name can produce another first-connection prompt.

SSH Security And Root Protection:

Chris used SSH configuration and root-access scenarios to explain why passwords, sudo, and private keys alone do not fully protect privileged access, and outlined layered controls for the cyber-defense club.

- **Root Exposure Risks:** Chris explained that allowing users to log in with passwords and then use sudo can leave root effectively exposed if the user's account is compromised. A separate root password or an SSH private key does not fully solve the problem if the attacker controls the session, because credentials can be captured or commands such as sudo can be replaced.
- **Layered Controls:** Chris described stronger approaches that move an additional authentication check outside the compromised session. Examples included separate administrative accounts and domains, hardware keys such as YubiKeys, short-lived certificates, password vaults, and temporary credentials that expire after a session.
- **Temporary Administrative Access:** Chris described a workplace model in which everyday accounts and privileged accounts reside in isolated domains. A security device authenticates the user with the everyday account, creates a temporary password for the privileged domain, and continuously changes it so the password is not reusable after logout.
- **SSH Configuration Files:** Chris showed that SSH configuration is stored under /etc/ssh, including the main configuration and configuration snippets in /etc/ssh/ssh_config.d or the corresponding server configuration directory. Files are read in lexical order, and for most settings the first value encountered wins; therefore, an earlier snippet can override a later setting in the main file.

Ubuntu VM Installation And Boot Media:

Chris helped a participant recover an incomplete Ubuntu VM by correcting the CD/DVD and ISO settings, explaining the installation choices, and clarifying how to boot from the installed disk afterward.

- **Removing Boot Media:** Chris found that the VM still had the Ubuntu installer attached, causing it to boot into the try-or-install screen. He instructed the participant to open Hardware, edit the CD/DVD drive, choose Do not use any media, apply the change, and reboot.
- **Reinstalling Ubuntu:** When the VM did not boot because no operating system was installed, Chris explained that the participant could either reinstall Ubuntu on the existing empty VM or create a new VM. The Ubuntu Desktop AMD64 ISO was selected and attached to the virtual CD/DVD drive.
- **Installation Choices:** Chris advised selecting Install Ubuntu, using the interactive installation, keeping the default software choices, and completing the user and time-zone prompts. The installation may take time, and the participant can disconnect and reconnect later to check progress.
- **Post-Installation Boot Order:** After installation, Chris instructed the participant to remove the ISO or set the CD/DVD drive to no media before rebooting. Otherwise, the VM may continue booting from the installer instead of the newly installed virtual hard disk.

Linux Administration Practice:

Chris demonstrated basic administrative tasks including package metadata updates, user creation, privilege considerations, display scaling, and locating SSH-related files.

- **APT Metadata Updates:** Chris clarified that sudo apt update refreshes package metadata and dependency information rather than upgrading installed packages. It should normally be run before apt install; the command may fail without sudo because APT needs access to its lock files.
- **Creating Users:** Chris demonstrated adduser craig, which creates the account, group, home directory, and prompts for a password and profile information. The command does not automatically make the user an administrator; administrative group membership must be configured separately.

- **Display Scaling:** Chris showed that Ubuntu display settings can be used to increase scaling, such as 150% or 200%, when a SPICE desktop is difficult to read. Fractional scaling is available but may reduce speed or display sharpness.
- **Recovery Access:** In response to a password-recovery question, Chris explained that Proxmox or Ubuntu recovery/single-user mode can provide a root shell without normal network startup, allowing an administrator to reset a password when ordinary login is unavailable.

Club Transition Planning:

Chris and the team discussed planning for Chris's retirement, continued club involvement, future meeting space, and a replacement for Teams.

- **Retirement Timeline:** Chris's countdown indicates retirement on January 4, 2027, and Chris clarified that retirement does not immediately end their involvement with the club; they expect to remain available at least through spring, although their future workload may change.
- **Future Meeting Space:** Chris and the team began planning for a possible change in meeting location around April. A library or another small-group venue was suggested, with internet access identified as an important requirement; portable 5G connectivity was mentioned as a possible fallback.
- **Transition Planning:** Chris proposed gradually transferring responsibilities between now and approximately April rather than waiting for a sudden handoff. The participants discussed clarifying who will continue participating and what level of support will be appropriate.
- **Meeting Platform:** Because the group expects to move away from Teams in January, Chris and the team considered Jitsi, including its integration with Zulip, as a possible replacement. The group noted that free Teams meetings may not support sessions of this length.

Follow-up tasks:

- **Ubuntu VM Installation:** Complete the Ubuntu installation on the VM and remove the installation DVD/media afterward so the VM boots from its hard disk. (Axel)

- **VM Networking:** Correct the VM network configuration so it uses the 144.76.90.x address range, the correct gateway and DNS/proxy settings, and verify connectivity afterward. (Axel, Naomi)
- **Proxmox CPU Configuration:** Change applicable VMs from the `host` CPU type to the recommended x86/AES CPU type while the VMs are shut down, then migrate any VM that remains on the overloaded host if needed. (the team)
- **Virt-Viewer Setup:** Install Virt-Viewer on the local Windows or Ubuntu computer where it is needed for VM console access. (the team)
- **SSH Service Verification:** Verify that the OpenSSH server is installed, enabled to start at boot, and active on each Linux VM. (the team)
- **SSH Configuration Homework:** Review the SSH settings assigned for next week and investigate the settings related to restricting console access and protecting root accounts. (the team)
- **Root Protection Guidance:** Add the root-protection guidance and the remaining SSH settings to the shared homework document or GitHub repository, then notify the team when it is available. (Chris)
- **CDC Security Recommendations:** Post or circulate the lightweight, low-cost security recommendations generated for the high-school CDC environment. (Chris)
- **Team Transition Planning:** Begin developing a transition plan covering the move away from Teams in January and identifying replacement meeting space and collaboration tools for the period after April. (Chris)